

Highlights

Gated Cluster-Pooled Sign Constraints for Multi-Output Sparse Regression

Artur Sepp, Mika Kastenholz

- Pool univariate signs across data-driven response clusters in sparse regression
- A closed-form noise-floor gate abstains when the pooled marginal signal is weak
- Sign consistency proved under a within-cluster sign-coherence condition
- Pooling a size- m cluster sharpens the sign-recovery rate by up to $\sqrt{m}$
- Sign-recovery gain over per-response derivation grows with cluster recoverability

Gated Cluster-Pooled Sign Constraints for Multi-Output Sparse Regression

Artur Sepp^{a,*}, Mika Kastenholz^a

^aLGT Bank, Quantitative Analytics, Bleicherweg 30, Zurich, 8002, Switzerland

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ABSTRACT

In sparse multi-output regression, a predictor's marginal sign for a response is informative, but unstable when estimated one response at a time. Existing univariate sign guidance acts on a single response and imposes the marginal sign without testing whether the signal exceeds the noise. We pool the univariate slope across a data-driven cluster of responses, impose the pooled sign as a constraint, and abstain from the constraint through a closed-form noise-floor gate. With known clusters, pooling recovers 99% of coefficient signs against 68% per-response in simulation. With discovered clusters the gain is smaller and grows with cluster recoverability. On yeast eQTL data the discovered clusters agree on the marginal sign in 93% of significant cells. We prove that the gated pooled signs are sign-consistent on the true support under a within-cluster sign-coherence condition. We leave the oracle limiting distribution as a conjecture. Pooling over a size- m cluster raises the gate's effective sample toward mT , sharpening the sign-recovery rate by up to a factor $\sqrt{m}$. The gate abstains rather than assert an unsupported sign, so the recovered structure is safe to use in models where a wrong sign reverses the decision.

1. Introduction

The sign of a coefficient is often its most interpretable feature. In a factor model for portfolio construction the sign of a loading says whether an asset hedges or amplifies a factor, which then sets the covariance and the allocation (Sepp, Ossa and Kastenholz, 2026). A sign that points the wrong way reverses that economic reading, so a method that asserts an unsupported sign is worse than one that abstains. In sparse multi-output regression, where N responses share M predictors, the marginal sign of a predictor is informative but unstable in any single response. Pooling the marginal evidence across similar responses stabilizes the sign, but only when the marginal signal is real and the responses truly share a direction. We make the pooling precise, gate it against noise, and prove that the pooled signs are consistent.

Four lines of work meet here. Chatterjee, Hastie and Tibshirani (2025) derive a univariate sign for each response and impose it as a constraint, response by response, with no pooling and no abstention. The cooperative LASSO of Chiquet, Grandvalet and Charbonnier (2012) couples related responses into sign-coherent groups through a convex penalty. The penalty enforces a shared sign but never derives one and never tests the signal strength. The multivariate sparse group LASSO of Li, Nan and Zhu (2015) selects groups in the coefficient matrix of a multi-response regression, again without a sign. The group LASSO of Yuan and Lin (2006) imposes group structure but derives no signs. The adaptive LASSO of Zou (2006), with the group oracle theory of Wang and Leng (2008), reweights the penalty by a

*Corresponding author

✉ artur.sepp@lgt.com (A. Sepp)

ORCID(s):

pilot estimate. Marginal screening removes uninformative predictors by a marginal test (Fan and Lv, 2008). The relaxed LASSO of Meinshausen (2007) sharpens a penalized fit by a select-then-reestimate split. We combine the univariate sign of Chatterjee et al. (2025) with a data-driven response clustering, pool the sign within a cluster, and gate it by a closed-form test in the screening spirit of Fan and Lv (2008). We derive and gate the sign before estimation. The cooperative LASSO instead couples signs inside the penalty and never abstains.

We develop a cluster-pooled univariate sign derivation for multi-output sparse regression. An elementary residual-sum-of-squares identity reduces the gate to a single matrix product. We prove that the gated pooled signs are sign-consistent on the true support in the sense of Zhao and Yu (2006), under a within-cluster sign-coherence condition. We leave the oracle limiting distribution as a conjecture. We show that pooling over a cluster of size m raises the gate's effective sample toward mT , sharpening the sign-recovery rate by up to a factor $\sqrt{m}$. The within-cluster residual correlation sets how much of the gain is realized. In simulation, pooling over known clusters recovers 99% of coefficient signs against 68% per-response. The gain with discovered clusters is smaller and grows with cluster recoverability. Where marginal screening keeps or drops a predictor, the gate keeps or withholds its sign, so the method declines to assert a sign that the data do not support. To our knowledge, the method is the first to derive a noise-floor-gated, cluster-pooled univariate sign for multi-output sparse regression.¹

Section 2 develops the method. Section 3 states the sign-consistency result and the pooling rate. Section 4 reports the simulation study. Section 5 applies the method to yeast eQTL data. Section 6 discusses the implications. The appendices give the closed-form identity, the formal conditions and proof, and the reproduction details.

2. Methodology

We develop the sign-derivation mechanism which assigns a sign to each predictor-response pair and uses the signs as hard constraints in a sparse multi-response regression. We derive each sign from a univariate slope pooled across the responses in a cluster, and we gate the sign with a closed-form noise-floor test. The pooling raises the per-cell sample size by the cluster size, so a shared sign rests on more evidence than a single response provides. We discover the clusters from the response correlation matrix, so the partition is data-driven rather than supplied.

Four lines of work meet in the mechanism. Chatterjee et al. (2025) derive per-response univariate signs and impose them as constraints, without pooling across responses. The group LASSO of Yuan and Lin (2006) imposes group structure on the coefficients but derives no signs. The adaptive LASSO of Zou (2006), with the group oracle theory of Wang and Leng (2008), reweights the penalty by a pilot estimate, again without sign derivation. Marginal screening removes uninformative predictors by a marginal test (Fan and Lv, 2008), but derives no sign. We pool the univariate

¹The method is implemented in the open-source Python package `factorlasso` (Sepp and Kastenholz, 2026b). The package and its interface are described in the companion software paper (Sepp and Kastenholz, 2026a).

sign across a discovered cluster, gate it by a test in the spirit of marginal screening, and prove sign consistency for the pooled and gated signs (Appendix B).

To our knowledge, the gated cluster-pooled univariate sign derivation is new among multi-response sparse-regression methods. The companion paper (Sepp and Kastenholz, 2026a) documents the software and comparison with peer methods.

Section 2.1 fixes notation. Section 2.2 derives the pooled slope. Section 2.3 states the gate. Section 2.4 describes the clustering. Section 2.5 states the constrained estimator. Appendix A derives the closed-form identity, and Appendix B states the sign-consistency theory.

2.1. Multi-response setup

Let $\mathbf{Y} \in \mathbb{R}^{T \times N}$ collect N responses observed over T observations, and let $\mathbf{X} \in \mathbb{R}^{T \times M}$ collect M predictors observed over the same observations. The multi-response regression model is

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta}^\top + \boldsymbol{\varepsilon}, \quad (1)$$

where $\boldsymbol{\beta} \in \mathbb{R}^{N \times M}$ is the coefficient matrix and $\boldsymbol{\varepsilon} \in \mathbb{R}^{T \times N}$ is the residual. The estimator centers the columns of $\mathbf{Y}$ and $\mathbf{X}$ before fitting and recovers the intercept afterward. We write $\mathbf{x}_j \in \mathbb{R}^T$ for the j -th predictor column, $\mathbf{y}_k \in \mathbb{R}^T$ for the k -th response column, and β_{kj} for the (k, j) entry. We partition the N responses into clusters $\{C_1, \dots, C_G\}$, where each $C_g \subseteq \{1, \dots, N\}$ is a set of response indices and $|C_g|$ its size. We write $C(k)$ for the cluster containing response k .

2.2. Cluster-pooled univariate sign derivation

We derive the sign of each coefficient from a univariate slope, pooled across the responses that share a cluster. For predictor j , the pooled no-intercept slope across all N responses solves the first-order condition $\mathbf{x}_j^\top \sum_{k=1}^N (\mathbf{y}_k - \hat{\beta}_j \mathbf{x}_j) = 0$, which gives

$$\hat{\beta}_j = \frac{\mathbf{x}_j^\top \mathbf{y}_{\text{sum}}}{N \mathbf{x}_j^\top \mathbf{x}_j}, \quad \mathbf{y}_{\text{sum}} \equiv \sum_{k=1}^N \mathbf{y}_k \in \mathbb{R}^T. \quad (2)$$

Equation (2) generalizes the single-response univariate slope of Chatterjee et al. (2025) to many responses, with the response sum $\mathbf{y}_{\text{sum}}$ in place of a single response. When the responses are grouped into clusters, we compute the slope against the cluster-sum response and broadcast it to every cluster member,

$$\hat{\beta}_{j,C} = \frac{\mathbf{x}_j^\top \mathbf{y}_{C,\text{sum}}}{|C| \mathbf{x}_j^\top \mathbf{x}_j}, \quad \mathbf{y}_{C,\text{sum}} = \sum_{k \in C} \mathbf{y}_k \in \mathbb{R}^T. \quad (3)$$

Equation (3) restricts Equation (2) to the cluster index set, and recovers it when $C = \{1, \dots, N\}$. The cluster-pooled slope is the OLS slope of the cluster-mean response $\bar{\mathbf{y}}_C = |C|^{-1} \mathbf{y}_{C,\text{sum}}$ on $\mathbf{x}_j$.

The pooled slope yields within-cluster sign coherence by construction. When two responses in a cluster are highly correlated, the pooled slope gives them one shared sign. The shared sign removes the within-cluster sign alternations that an unconstrained fit produces when it minimizes in-sample residual variance.

2.3. The noise-floor gate

We filter signs driven by sampling noise with a closed-form test. We compute the univariate t statistic from a residual sum-of-squares identity that avoids forming the per-cell residuals,

$$\text{SSR}_j = \|\mathbf{Y}\|_F^2 - N \hat{\beta}_j^2 (\mathbf{x}_j^\top \mathbf{x}_j), \quad (4)$$

which follows from the slope-defining identity $\mathbf{x}_j^\top \mathbf{y}_{\text{sum}} = N \hat{\beta}_j (\mathbf{x}_j^\top \mathbf{x}_j)$ (Appendix A). Equation (4) is one matrix product over j , so we compute the whole (N, M) sign matrix in a single pass. The pooled variance, standard error, and t statistic follow in closed form, and the cluster-pooled $t_{j,C}$ substitutes $|C|$ for N and restricts the Frobenius norm to the cluster columns (Appendix A). The gated sign matrix is

$$s_{kj} = \begin{cases} \text{sign}(\hat{\beta}_{j,C(k)}), & |t_{j,C(k)}| \geq \tau, \\ 0, & |t_{j,C(k)}| < \tau, \end{cases} \quad (5)$$

where τ is the threshold, with default $\tau = 0.75$. The threshold is not a significance test, because the pooled slopes carry no exact null distribution here. The threshold is a defensive filter that removes the worst noise-driven signs and keeps moderate-evidence signals. The gate operates at cluster granularity. When $|t_{j,C}| < \tau$ it sets $s_{kj} = 0$ for every member $k \in C$, which forces $\beta_{kj} = 0$ for the whole cluster on predictor j and preserves within-cluster sign coherence as a strict invariant.

2.4. Cluster discovery

The cluster partition can be supplied externally or discovered from the data. We discover it by Ward linkage on the response correlation matrix. Let $\mathbf{C} = \text{corr}(\mathbf{Y})$ be the $N \times N$ correlation matrix of the responses. We form the dissimilarity $\mathbf{D} = \mathbf{1} - \mathbf{C}$, apply Ward agglomerative linkage, and cut the dendrogram at height $\rho_{\text{cut}} \cdot \max_{i,j} D_{ij}$, where $\rho_{\text{cut}} \in (0, 1]$ is the cutoff fraction, with default 0.5. We use $1 - \rho_{ij}$ as the dissimilarity rather than the chord distance $\sqrt{2(1 - \rho_{ij})}$ between standardized response vectors. The two agree up to a concave monotone transform, so we read

the step as a stable correlation-clustering heuristic rather than exact Ward variance minimization.² Ward linkage is a standard, stable default for correlation-based agglomerative clustering. The discovered partition ranges from one cluster (at $\rho_{\text{cut}} \rightarrow 1$) to N singleton clusters (at $\rho_{\text{cut}} \rightarrow 0$), and is set by the data rather than chosen by the user.

2.5. Sign-constrained estimation

The derived signs constrain a multi-response sparse-group LASSO estimator (Simon, Friedman, Hastie and Tibshirani, 2013). The sign matrix $\mathcal{S} = (s_{kj}) \in \{-1, 0, +1, \text{NaN}\}^{N \times M}$ from Equation (5) defines the constraint set

$$\mathcal{K}(\mathcal{S}) = \left\{ \boldsymbol{\beta} : \beta_{kj} = 0 \text{ if } s_{kj} = 0, \quad s_{kj} \beta_{kj} \geq 0 \text{ if } s_{kj} \in \{-1, +1\} \right\}. \quad (6)$$

A gated-out cell is pinned to zero, and a signed cell is held in the matching half-space. Cells with $s_{kj} = \text{NaN}$ are unconstrained. The estimator solves the prior-centered sparse group LASSO problem

$$\begin{aligned} \hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathcal{K}(\mathcal{S})} & \left\{ \frac{1}{T} \sum_{t=1}^T \left\| \mathbf{y}_t - \boldsymbol{\beta} \mathbf{x}_t \right\|_2^2 + \lambda (1 - \alpha) \sum_{g=1}^G \sqrt{|C_g|/G} \sum_{k \in C_g} \left\| \boldsymbol{\beta}_{k,\cdot} - \boldsymbol{\beta}_{k,\cdot}^0 \right\|_2 \right. \\ & \left. + \lambda \alpha \sum_{k=1}^N \sum_{j=1}^M W_{kj} \left| \beta_{kj} - \beta_{kj}^0 \right| \right\}, \end{aligned} \quad (7)$$

where $\lambda > 0$ is the penalty strength, $\alpha \in [0, 1]$ mixes the group and cell penalties, $\boldsymbol{\beta}^0$ is an optional target matrix (zero by default), the multiplier $\sqrt{|C_g|/G}$ moderates the influence of large clusters, and the cell weights $W_{kj} > 0$ (Equation (8), with $W_{kj} = 1$ when adaptive reweighting is off) reweight the cell penalty. The cell weights enter only the cell term, so adaptive reweighting is inactive when $\alpha = 0$.

The group penalty in Equation (7) runs the ℓ_2 norm over the loadings of one response, which we call the hierarchical-clustering group LASSO (HCGL) mode. A complementary mode runs the group norm over the responses of a cluster on one predictor, which we call the factor-clustering group LASSO (FCGL) mode. When adaptive reweighting is on, the penalty weights follow the adaptive LASSO of Zou (2006),

$$W_{kj} = \frac{1}{\max(|\hat{\beta}_{j,C(k)}|, \varepsilon)^\gamma}, \quad (8)$$

with the cluster-pooled slope of Equation (3) as the pilot, exponent $\gamma > 0$, and a stabilizer $\varepsilon > 0$ that prevents weight explosion on near-zero slopes. The companion paper (Sepp and Kastenholz, 2026a) gives the full cone solver, the FCGL penalty, the row- and cluster-aggregated adaptive weights, and the practitioner overlay that can pierce a cluster decision at the cell level.

²Ward's criterion is derived for Euclidean distances between points, so under $1 - \rho$ the merge costs are not exact within-cluster sum-of-squares increments. We retain $1 - \rho$ because the partition is used only to pool signs and to group the penalty, not to make a metric claim.

3. Sign consistency and the pooling rate

We prove that the gated cluster-pooled signs match the true signs on the support as the sample grows. We also quantify the rate at which pooling delivers this match. Let $\hat{\mathbf{S}}_T$ be the gated sign matrix at sample size T , and let $\mathbf{S}_{\text{true}}$ be the matrix of true signs. Proposition 1 states sign consistency on the support, in the sense of Zhao and Yu (2006). Conditional on the matched signs, the constrained estimator recovers the true support (Appendix B).

Pooling sets the rate. The gate statistic for a predictor in a cluster of size m uses the closed-form standard error of Equation (10), which pools the m cluster responses with the T observations. With independent within-cluster residuals the effective sample size is mT , so the support statistic diverges at the $\sqrt{mT}$ rate, a factor $\sqrt{m}$ above the per-response $\sqrt{T}$ rate. Correlated residuals shrink the gain. For a common within-cluster residual correlation $\bar{\rho}$, the design effect $1 + (m-1)\bar{\rho}$ reduces the effective sample to $mT / (1 + (m-1)\bar{\rho})$ and the realized rate to $\sqrt{m / (1 + (m-1)\bar{\rho})}$. The realized rate equals $\sqrt{m}$ as $\bar{\rho} \rightarrow 0$ and falls to one as $\bar{\rho} \rightarrow 1$. The gain is therefore largest as $\bar{\rho} \rightarrow 0$. This residual correlation is distinct from the total correlation that the clustering reads, so a coherent cluster need not carry a large $\bar{\rho}$. Figure 3 plots the $\sqrt{mT}$ ideal, which the realized recovery approaches from below.

Two conditions carry the result. The cluster sign coherence condition asks the discovered clusters to respect the true within-cluster sign structure, which is weaker than recovering the true clusters. The marginal sign agreement condition asks the population pooled slope to share the sign of the partial coefficient. The second condition holds when the predictors are uncorrelated, and it can fail under strong cross-predictor leakage of opposing sign. The mixed-sign regimes of Section 4 probe both conditions.

The result is about the support, not the null cells. At a fixed threshold the gate retains a non-vanishing fraction of truly-null cells, so the gate is a screen rather than a model-selection oracle. Under the per-response gate this fraction is about $2\Phi(-\tau) \approx 0.45$ at the default $\tau = 0.75$. Under cluster pooling the retained fraction runs higher, because the pooled variance estimate does not adjust for within-cluster dependence. A retained constraint on a null cell is a one-sided half-space whose boundary contains zero, so it cannot exclude the value that the penalty drives the cell toward. Exact recovery of every sign would require a diverging threshold. The formal conditions, the oracle-distribution conjecture, and the proof are in Appendix B.

4. Simulation study

We use simulation to locate where cluster-pooled sign derivation recovers coefficient signs better than a per-response derivation, and to show how the gate controls false signs. The mechanism is direct. Pooling the univariate slope across the responses in a cluster raises the per-cell sample size by the cluster size, so a shared sign decision rests on more evidence than any single response provides. The gain therefore depends on whether the responses truly share

a sign and on whether the clusters can be recovered from the data. We vary both, together with the sample size, and measure how often each derivation returns the population sign of an active coefficient.

4.1. Design

We generate data from a multi-response factor model with known cluster structure. We draw T observations on N responses and M predictors, and organize the responses into G equal-size clusters. Within a cluster, the responses share the same active predictors and the same coefficient signs, and we draw the coefficient magnitudes independently from the uniform distribution on $[0.5, 1.5]$. We draw the predictor columns independently from the standard normal, so a null coefficient carries no population marginal association and the sign target on null cells is unambiguous. We scale the residual variance so each response attains a target population R^2 . The recovery target is the population sign $\text{sign}(\beta_{kj})$ on the active cells, where $\beta_{kj} \neq 0$.

We compare three ways of deriving the sign matrix $\mathbf{S}$, holding the gate fixed. The per-response derivation computes each response's sign from its own univariate slope, with no pooling, following the single-response rule of Chatterjee et al. (2025). Known-cluster pooling computes the cluster-pooled slope of Equation (3) from the true partition. Estimated-cluster pooling discovers the partition by Ward linkage on the response correlation matrix, then pools within the discovered clusters, the HCGL derivation of Section 2.4. All three pass the pooled slope through the gate of Equation (5).

We report four rates. Sign recovery is the fraction of active cells whose gated sign equals the population sign. The sign-flip rate is the fraction of active cells assigned the opposite nonzero sign. The abstention rate is the fraction of active cells the gate sets to zero. These three rates partition the active cells. The false-sign rate is the fraction of null cells assigned any nonzero sign, and we report it separately. In the base configuration we set $N = 60$, $M = 20$, $G = 6$, $T = 80$, five active predictors per response, and population $R^2 = 0.10$. We fix the gate threshold at the package default $\tau = 0.75$ unless stated otherwise, and we report Monte Carlo means with standard errors, using the replication count given in each caption.

4.2. Sign recovery and the recoverability boundary

Known-cluster pooling nearly removes abstention on true signals. Table 1 reports the base configuration. Known-cluster pooling recovers the population sign in 0.993 of active cells, against 0.683 for the per-response derivation. The gain comes from abstention rather than from flips. The per-response derivation abstains on 0.287 of active cells, while known-cluster pooling abstains on 0.007. Pooling raises the per-cell sample size by the cluster size, so the gate clears far more true signals at the same threshold. Clearing a fixed threshold more often reflects the larger effective sample under pooling, so the recovery rates should be read together with the false-sign rates they carry. Estimated-cluster

Table 1

Known-cluster pooling nearly eliminates abstention on true signals. Monte Carlo sign-recovery, sign-flip, abstention, and false-sign rates for the three sign derivations in the base configuration ($N = 60$ responses, $M = 20$ predictors, $G = 6$ clusters, $T = 80$ observations, population $R^2 = 0.10$, gate threshold $\tau = 0.75$). The first three rates are computed on active coefficients and the false-sign rate on null coefficients. Cells are means over 200 replications, with standard errors in parentheses.

Method	Sign recovery	Sign flip	Abstain (true)	False sign
Per-response (LASSO)	0.683 (0.002)	0.030 (0.001)	0.287 (0.002)	0.459 (0.001)
Pooled, known clusters	0.993 (0.001)	0.000 (0.000)	0.007 (0.001)	0.579 (0.004)
Pooled, estimated (HCGL)	0.697 (0.002)	0.032 (0.001)	0.271 (0.002)	0.479 (0.001)

pooling recovers 0.697, close to the per-response rate, because at $R^2 = 0.10$ the response correlation matrix does not yet reveal the clusters.

Pooling carries a cost on null cells. The false-sign rate is 0.579 for known-cluster pooling against 0.459 for the per-response derivation (Table 1). The per-response rate matches the $2\Phi(-\tau) = 0.453$ that an asymptotically standard-normal gate statistic predicts, while pooling runs higher because its pooled variance estimate treats the within-cluster responses as independent and so over-disperses the null statistic (Appendix B). The gate threshold controls this rate directly, as the next subsection shows.

The advantage of estimated-cluster pooling grows with cluster recoverability. Figure 1 traces sign recovery as the population R^2 rises from 0.05 to 0.50 at fixed $T = 80$. Known-cluster pooling recovers between 0.960 and 1.000 across the whole range, because the oracle partition does not depend on the data. Estimated-cluster pooling matches the per-response derivation at $R^2 = 0.05$ (0.558 against 0.552) and pulls ahead as R^2 rises, reaching 0.996 against 0.954 at $R^2 = 0.50$. The gap between estimated-cluster pooling and the per-response derivation measures how clearly the data reveal the clusters. The gap is near zero when the clusters are invisible and widens as they emerge.

Pooling helps most when the sample is small relative to the structure. Table 2 crosses the population R^2 with the sample size T . The gap reaches 0.069 at $R^2 = 0.30$ and $T = 60$ (0.919 against 0.850). The gap closes as T grows, falling to 0.003 at $T = 240$, because the per-response derivation becomes consistent on its own once the sample is large. The practical reading is that pooling earns its keep in the scarce-sample regime, and adds little when the per-response signs already clear the gate.

4.3. The noise-floor gate

The threshold τ trades recovered true signs for false-sign control along a smooth frontier. Figure 2 plots sensitivity against the false-sign rate as τ varies from 0.5 to 3.5, under known-cluster pooling. At $\tau = 0.75$ the gate recovers 0.993 of true signs but admits a false-sign rate of 0.579. At $\tau = 2.5$ it recovers 0.863 of true signs at a false-sign rate of 0.070. The frontier bends near $\tau = 2.0$ to 2.5, which we read as the useful operating range for a final sign report. The package

Table 2

Pooling's advantage is largest when structure is recoverable and the sample is small. Sign recovery for estimated-cluster pooling (HCGL) and per-response derivation across population R^2 and sample size T , with their difference (Gap). All other settings as in the base configuration. Cells are means over 200 replications.

R^2	T	Pooled (HCGL)	Per-response	Gap
0.10	60	0.646	0.624	+0.022
0.10	120	0.765	0.764	+0.001
0.10	240	0.887	0.887	+0.000
0.30	60	0.919	0.850	+0.069
0.30	120	0.969	0.945	+0.024
0.30	240	0.989	0.986	+0.003
0.50	60	0.992	0.925	+0.066
0.50	120	0.999	0.980	+0.019
0.50	240	1.000	0.997	+0.003

default of 0.75 is permissive by design, since it suits a screening step that removes the worst noise-driven constraints before regularization.

The over-dispersion is the design effect, which can be removed. The pooled standard error divides by the cluster size as if the responses were independent, which omits the factor $1 + (|C| - 1)\bar{\rho}$. A design-effect-corrected gate scales the standard error by the square root of this factor. The corrected gate brings the known-cluster false-sign rate from 0.579 to 0.457, matching the per-response $2\Phi(-\tau) = 0.453$, at a recovery cost from 0.993 to 0.988 (replication code). At this matched false-sign rate near 0.45, known-cluster pooling recovers 0.988 against 0.683 for the per-response derivation, so the recovery advantage does not come from the higher false-sign rate. The correction leaves the flip rate unchanged, because the marginal-versus-partial disagreement of Section 4.5 is a bias rather than a variance. We keep the uncorrected gate as the default screen before regularization, and recommend the design-effect-corrected gate when the signs are reported as a final, calibrated result.

The design effect also bounds the realized pooling gain. The base design draws independent within-cluster residuals, for which $\bar{\rho} \approx 0$. We correlate the residuals within each cluster and sweep $\bar{\rho}$, holding the base configuration fixed (Table 3). As $\bar{\rho}$ rises, the known-cluster recovery falls from 0.990 to 0.863 and the false-sign rate rises from 0.577 to 0.809. The design effect lowers the pooled effective sample below mT , which drives both moves. The realized rate factor falls from $\sqrt{m}$ toward one, so the gain is largest when the cluster carries shared signal with weakly correlated residuals.

The threshold also sets a multiplicity budget. The gate screens $N \times M$ cells. At threshold τ the gate admits a null cell with probability about $2\Phi(-\tau)$. The expected number of false constraints is therefore about $2\Phi(-\tau)NM$, which the default $\tau = 0.75$ leaves large and $\tau = 2.5$ controls. A user with a target false-sign rate reads τ from the frontier of Figure 2. A user with a per-family budget sets $2\Phi(-\tau)NM$ to the tolerated count.

Table 3

The pooling gain attenuates as within-cluster residual correlation rises. Known-cluster sign recovery, abstention, and false-sign rate against the within-cluster residual correlation $\bar{\rho}$, in the base configuration ($N = 60$, $M = 20$, $G = 6$, $T = 80$, $R^2 = 0.10$, $\tau = 0.75$, cluster size $m = 10$). The rate factor $\sqrt{m/(1 + (m-1)\bar{\rho})}$ is the realized multiple of the per-response convergence rate against the ideal $\sqrt{m} = 3.16$. Means over 100 replications.

$\bar{\rho}$	Recovery	Abstention	False-sign	Rate factor
0.0	0.990	0.010	0.577	3.16
0.3	0.939	0.050	0.718	1.64
0.6	0.898	0.067	0.777	1.25
0.9	0.863	0.082	0.809	1.05

The discovered clusters are stable across the cutoff fraction. Sign recovery for estimated-cluster pooling runs from 0.683 at $\rho_{\text{cut}} = 0.3$ to 0.724 at $\rho_{\text{cut}} = 0.7$, with no cliff. A low cutoff merges few responses and reduces to the per-response derivation, while a high cutoff pools more aggressively and trades false signs for recovery. The default $\rho_{\text{cut}} = 0.5$ sits at the midpoint of this range.

4.4. Sign consistency

Pooling reaches sign consistency at a far smaller sample. Figure 3 shows sign recovery against the sample size T at fixed $R^2 = 0.10$. Both derivations are sign-consistent, since recovery rises toward one as T grows. Known-cluster pooling reaches 1.000 by $T = 160$, while the per-response derivation needs $T = 640$ to reach 0.979. Pooling attains consistency at roughly a quarter of the sample size. Appendix B states the rate that this curve illustrates.

The same contrast is visible cell by cell. Figure 4 shows the three sign matrices for one replication at $R^2 = 0.20$ and $\tau = 2.0$. The known-cluster matrix reproduces the population signs as coherent within-cluster blocks. The per-response matrix is fragmented, with isolated cells and within-cluster sign alternations on the active predictors. Pooling removes the alternations by sharing one decision across each cluster.

4.5. Correlated predictors and the marginal-sign assumption

The method derives a sign from a marginal slope but imposes it on a partial coefficient. Condition (A6) of Appendix B asks the two to agree, which holds when the predictors are uncorrelated. The base design draws independent predictors, so (A6) holds by construction. We now correlate the predictors to stress it.

We draw the factors with equicorrelation ρ_X and sweep ρ_X from 0 to 0.8, holding the rest of the base configuration fixed. Figure 5 reports the three sign rates. As ρ_X rises, cross-predictor leakage pulls the marginal slope away from the partial coefficient. The derived sign then flips. The sign-flip rate rises for every derivation. Pooling amplifies it. At $\rho_X = 0.8$ the known-cluster flip rate is 0.296 against 0.217 for the per-response derivation. The false-sign rate saturates near one, because correlated predictors give null cells a spurious marginal association.

Table 4

The hard sign constraint never asserts a wrong sign, while the cooperative LASSO reaches higher recovery and lower coefficient error by flipping 10% to 22% of signs. Sign recovery, flip rate, and abstention on active cells, with the coefficient mean squared error (β -MSE), for a hard sign-constrained fit and the cooperative LASSO of Chiquet et al. (2012), both on known clusters, against the predictor equicorrelation ρ_X , in the base configuration. Each penalty minimizes β -MSE, and the sign is read from the fitted coefficient. Means over 40 replications.

ρ_X	Hard sign constraint				Cooperative LASSO			
	Rec.	Flip	Abst.	β -MSE	Rec.	Flip	Abst.	β -MSE
0.0	0.743	0.000	0.257	0.0044	0.895	0.101	0.004	0.0026
0.2	0.598	0.002	0.400	0.0046	0.895	0.099	0.005	0.0028
0.4	0.487	0.005	0.508	0.0053	0.889	0.101	0.010	0.0032
0.6	0.257	0.006	0.738	0.0071	0.857	0.131	0.012	0.0044
0.8	0.109	0.004	0.887	0.0109	0.771	0.220	0.009	0.0084

A wrong marginal sign is more costly than an abstention. The constraint forces the partial coefficient into the wrong half-space, an asymmetric and confident error. Pooling sharpens the wrong sign as readily as the right one. The recovery advantage of known-cluster pooling narrows sharply as ρ_X rises, from 0.31 at $\rho_X = 0$ to 0.05 at $\rho_X = 0.8$. Above $\rho_X \approx 0.5$ the pooled flip rate exceeds the per-response rate, so pooling trades a small recovery edge for more confident errors. Cluster-pooled sign derivation suits weakly correlated predictors. Under strong predictor correlation the marginal sign becomes unreliable. A soft penalty that the data can overrule then recovers more signs, at the cost of asserting more wrong ones.

We benchmark the hard sign constraint against the cooperative LASSO of Chiquet et al. (2012), a soft penalty that couples the signs of a cluster but lets the data overrule them. We fit both on the known clusters across the same ρ_X grid and select each penalty to minimize coefficient error. We read the sign from the fitted coefficient (Table 4), not from the derived sign of Figure 5. A wrong derived sign holds the coefficient in the wrong half-space, where the penalty then drives it to zero. The fitted cell reads as an abstention rather than a flip. The two methods make different kinds of error. The hard constraint never asserts a wrong sign, holding its flip rate at 0.006 or below across the grid. It abstains instead, and its abstention reaches 0.887 at $\rho_X = 0.8$ as the marginal signal becomes unreliable. The cooperative LASSO commits to a sign on almost every cell, so it asserts a wrong sign on 0.101 to 0.220 of cells as leakage grows. It recovers more signs in exchange, from 0.895 to 0.771 across the grid, against 0.743 to 0.109 for the hard constraint. It also reaches lower coefficient error throughout. An abstention assigns no sign, but a flip pins a coefficient to the wrong half-space. For a method whose output is a sign, the two are not the same error.

The eQTL application of Section 5 sits below the adversarial regime. The screened markers retain some linkage correlation. We collapse tight-linkage runs to reduce it, so the design correlation is modest rather than extreme.

These results locate the value of cluster-pooled sign derivation. Pooling helps when the responses carry recoverable cluster structure and the sample is too small for per-response signs to clear the gate. Pooling adds little when the sample is large or the clusters are invisible. Pooling can hurt when the predictors are strongly correlated, because it sharpens

a flipped marginal sign into a wrong constraint. The threshold τ , not the pooling, governs the false-sign rate, so a user tunes the two concerns separately. A user with a known grouping gains the most, and a user relying on discovered clusters gains in proportion to how clearly the data reveal them.

5. Application to yeast eQTL data

We apply the method to a yeast expression quantitative trait locus (eQTL) data set, where the responses are gene expression levels and the predictors are genetic markers. The application illustrates the method on real data. We examine whether real responses carry the sign-coherent cluster structure that the method exploits. We then read the recovered structure against known biology. We choose a data set that sits in the low-recoverability regime of Section 4, so we report interpretable structure rather than a prediction gain.

5.1. Data and setup

The data come from a cross of two yeast strains (Brem and Kruglyak, 2005), with $T = 112$ segregants. We study the genes of the mitogen-activated protein kinase (MAPK) signaling pathway, which divides into four sub-pathways, namely the pheromone response, filamentation, high-osmolarity response, and cell-wall integrity. We take the KEGG pathway membership measured in the data, which gives $N = 64$ genes.³ We reduce the marker panel from 3244 markers to representative markers by collapsing consecutive runs in tight linkage, then keep the markers associated with at least two genes at $p \leq 0.01$, the screen of Yin and Li (2011). The screen leaves $M = 202$ markers. We standardize the genes and the markers, and fit estimated-cluster pooling at cutoff $\rho_{\text{cut}} = 0.7$ and threshold $\tau = 2.0$, within the operating range identified in Section 4.3.

5.2. Recovered clusters and sign coherence

The discovered clusters are sign-coherent. The method recovers nine gene clusters. Within each cluster, the genes agree on the marginal sign of their marker associations. The agreement reaches 0.93 of the cells that carry a significant association at $p \leq 0.05$ (Table 5). The agreement is a within-cluster consistency check, not independent validation. The method forms the clusters from the same correlations that drive the shared sign, then computes the agreement on the significant cells alone. The agreement still shows that the discovered groups carry a single sign direction on these data. The per-cluster agreement ranges from 0.84 to 1.00. The residual correlation within the clusters, after removing the marker signal, averages 0.22, against a total within-cluster correlation of 0.36. The residual correlation governs the realized pooling gain through the design effect of Section 4.3, and it runs below the total correlation that drives the clustering. At the typical cluster size, the clusters realize about 1.7 times the per-response convergence rate, below the ideal $\sqrt{m}$ of 2.7.

³We use the KEGG MAPK membership measured in the data for reproducibility. The conditional-Gaussian-graphical-model literature (Yin and Li, 2011) works with a canonical 54-gene MAPK subset. We keep the full measured membership here.

Table 5

Recovered gene clusters are sign-coherent within, and partly aligned to the MAPK sub-pathways. For each discovered cluster: the number of genes, the dominant sub-pathway with its purity, the number of markers with a nonzero loading on any cluster gene, and the within-cluster raw marginal-sign agreement. Yeast eQTL cross of Brem and Kruglyak (2005): 64 MAPK genes, 202 markers, 112 segregants. Estimated-cluster pooling at $\rho_{\text{cut}} = 0.7$ and $\tau = 2.0$.

Cluster	Genes	Dominant sub-pathway	Purity	Markers	Sign coherence
1	5	Hog1	0.60	59	0.92
2	8	Hog1	0.50	57	0.84
3	4	Slr2	0.50	51	0.89
4	7	Kss1	0.43	62	0.97
5	7	Hog1	0.71	120	1.00
6	6	Slr2	0.50	86	0.95
7	13	Slr2	0.38	79	0.90
8	4	Fus3	1.00	41	1.00
9	10	Fus3	0.90	116	0.94

The clusters track the sub-pathways only in part. The adjusted Rand index between the clusters and the four sub-pathways is 0.114, and the mean cluster purity is 0.614 (Table 5). The two pheromone-response clusters are clean, at purity 1.00 and 0.90, while the cell-wall and osmolarity genes mix across clusters. Co-expression groups the genes by shared regulation, which need not follow the pathway annotation. Figure 6 orders the gene correlation matrix by cluster and colors each gene by its sub-pathway.

5.3. Genomic structure

The gated signs form coherent blocks along the genome. Figure 7 shows the gated sign matrix, with genes ordered by cluster and markers ordered along the genome. The signs form coherent horizontal blocks within each cluster, the structure that the agreement statistic measures. The gate leaves many cells at zero, where the pooled evidence falls below τ .

The marker loadings concentrate at a few genomic positions. Figure 8 counts, for each marker, the genes that load on it. A locus on chromosome 14 loads 42 of the 64 genes, and loci on chromosomes 12, 3, and 5 each load more than 30. The peaks recover the trans-acting eQTL hotspots that regulate many genes at once, a documented feature of this cross (Brem and Kruglyak, 2005).

5.4. Prediction

Pooling does not improve prediction on these data. In three-fold cross-validation the per-response LASSO attains a median out-of-sample R^2 of 0.118 per gene, against 0.088 for the factor-cluster variant and 0.038 for HCGL. The ranking matches the simulation. At the recoverability of this data set, pooling sign constraints over weakly separated clusters constrains the fit without a prediction payoff. The contribution here is the coherent, gated sign structure rather than the forecast.

The discovered clusters are sign-coherent and map onto interpretable genomic hotspots. The lower out-of-sample prediction marks the boundary of the method. A user adopts cluster-pooled sign derivation for coherent, interpretable structure in the small-sample regime, and not for a forecast gain where per-response signs already suffice.

6. Discussion

We develop a sign-derivation mechanism for multi-output sparse regression and establish when it helps. The mechanism pools the univariate sign across a data-driven cluster and gates it by a closed-form test. The pooling raises the per-cell sample size by the cluster size. In the base simulation the pooling recovers the population sign in 99.3% of active cells against 68.3% for the per-response derivation. The gain comes almost entirely from clearing true signals that the per-response gate abstains on, a 0.7% abstention rate against 28.7%.

The gain depends on the regime. The advantage over per-response derivation tracks cluster recoverability, and is largest when the sample is small relative to the structure, with a gap of 0.069 at population $R^2 = 0.30$ and $T = 60$. The gain vanishes as the sample grows, because the per-response signs become consistent on their own. The threshold governs the false-sign rate along a smooth frontier, with a sensitivity of 0.86 at a false-sign rate of 0.07 at $\tau = 2.5$, so a user separates the pooling decision from the false-sign tolerance. Pooling reaches sign consistency at roughly a quarter of the sample the per-response derivation needs.

On yeast eQTL data the recovered gene clusters agree on the marginal sign in 93% of significant cells, so the structure the method pools over is present in real data. The cluster-marker loadings concentrate at a few genomic positions, the known trans-acting hotspots of the cross. Pooling does not improve prediction here, at a median out-of-sample R^2 of 0.038 against 0.118 for the per-response LASSO, which the low recoverability of the data predicts. The contribution on this data set is interpretable sign structure rather than forecast accuracy.

The mechanism changes how a practitioner derives sign constraints. A practitioner with a known grouping gains the most. Factor models for portfolio optimization are one such setting, where the factor grouping is given. In that setting, sign recovery keeps the estimated factor model economically tractable, which the strategic and tactical allocation built on it requires (Sepp et al., 2026). A practitioner relying on discovered clusters gains in proportion to how clearly the data reveal them, and should question the marginal sign agreement condition first when the predictors carry strong cross-predictor correlation of opposing sign. For the literature, the result extends univariate sign guidance from a single response to a data-driven cluster, and pairs it with a gate that treats the sign as a screen rather than an unconditional constraint.

Two limitations remain. The oracle distribution of the constrained estimator is stated as Conjecture 1 and not proved. The marginal-sign-agreement condition fails under correlated predictors, where Section 4.5 shows the derived sign flips and pooling amplifies it. In that regime a cooperative LASSO, which the data can overrule, recovers more of

the signs at lower coefficient error than the hard constraint. The cost is a higher flip rate (Section 4.5). We benchmark against the per-response derivation, the cluster-size-one special case, and the cooperative LASSO of Chiquet et al. (2012).

A. The closed-form SSR identity

This appendix derives the residual sum-of-squares identity of Equation (4) and gives the full t statistic.

Derivation. For predictor j , the pooled no-intercept slope $\hat{\beta}_j$ of Equation (2) satisfies the first-order condition $\mathbf{x}_j^\top \mathbf{y}_{\text{sum}} = N \hat{\beta}_j (\mathbf{x}_j^\top \mathbf{x}_j)$. The pooled residual sum of squares aggregates the squared residuals of the N per-response fits at the common slope $\hat{\beta}_j$,

$$\begin{aligned} \text{SSR}_j &= \sum_{k=1}^N \left\| \mathbf{y}_k - \hat{\beta}_j \mathbf{x}_j \right\|_2^2 = \sum_{k=1}^N \left(\|\mathbf{y}_k\|_2^2 - 2 \hat{\beta}_j \mathbf{x}_j^\top \mathbf{y}_k + \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{x}_j \right) \\ &= \|\mathbf{Y}\|_F^2 - 2 \hat{\beta}_j \mathbf{x}_j^\top \mathbf{y}_{\text{sum}} + N \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{x}_j, \end{aligned} \quad (9)$$

using $\sum_k \|\mathbf{y}_k\|_2^2 = \|\mathbf{Y}\|_F^2$ and $\sum_k \mathbf{x}_j^\top \mathbf{y}_k = \mathbf{x}_j^\top \mathbf{y}_{\text{sum}}$. Substituting the first-order condition $\mathbf{x}_j^\top \mathbf{y}_{\text{sum}} = N \hat{\beta}_j (\mathbf{x}_j^\top \mathbf{x}_j)$ into the middle term collapses Equation (9) to Equation (4), $\text{SSR}_j = \|\mathbf{Y}\|_F^2 - N \hat{\beta}_j^2 (\mathbf{x}_j^\top \mathbf{x}_j)$. The right-hand side requires only $\|\mathbf{Y}\|_F^2$ (computed once), the per-predictor inner product $\mathbf{x}_j^\top \mathbf{x}_j$, and the slope, so the full (N, M) statistic is one matrix product rather than a loop that forms the $T \times N$ residual matrix for each predictor.

Variance, standard error, and t statistic. The pooled variance estimator, standard error, and t statistic are

$$\hat{\sigma}_j^2 = \frac{\text{SSR}_j}{N T - N}, \quad \widehat{\text{SE}}(\hat{\beta}_j) = \sqrt{\frac{\hat{\sigma}_j^2}{N \mathbf{x}_j^\top \mathbf{x}_j}}, \quad t_j = \frac{\hat{\beta}_j}{\widehat{\text{SE}}(\hat{\beta}_j)}. \quad (10)$$

The cluster-pooled variant $t_{j,C}$ used in the gate of Equation (5) substitutes $|C|$ for N throughout Equation (10) and restricts the Frobenius norm to the cluster columns, $\sum_{k \in C} \|\mathbf{y}_k\|_2^2$. Missing entries are handled by per-cell validity-mask accounting, which reduces to Equation (10) when the data are fully observed (Sepp and Kastenholz, 2026a).

B. Conditions and proof for the sign-consistency result

This appendix states the support-sign consistency of the cluster-pooled sign-derivation mechanism and gives a proof sketch. The sign matrix is derived identically under the HCGL and FCGL modes, which gate the same cluster-pooled slope at the same threshold and differ only in the group norm of the penalty, so the argument applies to both. We establish the support-sign statement (Proposition 1) under conditions (A1)–(A6). The oracle distributional statement that would accompany a full theorem, namely asymptotic normality on the true support with OLS-equivalent

covariance, is recorded as Conjecture 1 and not proved here. The argument holds under stylized conditions rather than as a guarantee for every regime. Condition (A6) below requires the marginal cluster-pooled slope to agree in sign with the partial true coefficient, and strong cross-predictor collinearity of opposing sign is exactly what breaks it. Under such collinearity the asymptotic argument is not claimed to hold, and the evidence for the mechanism in that regime is the simulation of Section 4, not Proposition 1.

We adopt the notation of Section 2. Let $\beta_{\text{true}} \in \mathbb{R}^{N \times M}$ denote the true coefficient matrix in the data-generating model $Y = X \beta_{\text{true}}^\top + \epsilon$, and let $S_{\text{true}} \in \{-1, 0, +1\}^{N \times M}$ be the matrix of true signs, with $S_{\text{true},kj} = 0$ where $\beta_{\text{true},kj} = 0$. Let $\hat{S}_T$ denote the gated cluster-pooled sign matrix of Equations (3)–(5) at sample size T , and let $\hat{\beta}_T$ denote the constrained adaptive estimator of Equation (7) under $\hat{S}_T$ with the adaptive weights of Equation (8).

Regularity conditions.

(A1) Sub-Gaussian noise. The residual matrix ϵ has independent rows with mean zero and finite variance, and each entry ϵ_{ik} is sub-Gaussian with proxy variance bounded above by a fixed constant $\sigma_{\max}^2$.

(A2) Design regularity. The predictor matrix X satisfies the moment condition $T^{-1} X^\top X \rightarrow \Sigma_F$ as $T \rightarrow \infty$ with Σ_F positive definite, and the columns of X are bounded in probability.

(A3) Cluster sign coherence. The HCGL partition recovered from $\text{corr}(Y)$ respects the true within-cluster sign structure in the following sense. For every cluster $\hat{C}$ discovered on the sample (X, Y) and every predictor j , the signs $\{S_{\text{true},kj} : k \in \hat{C}, S_{\text{true},kj} \neq 0\}$ share a single value. The condition is weaker than perfect cluster recovery, because the discovered partition need not match the true cluster identity, only respect the true sign structure.

(A4) Separation. There exists a positive constant c^* such that, for every cluster C and every predictor j with $\beta_{\text{true},kj} \neq 0$ for some $k \in C$, the cluster-pooled population slope $\beta_{j,C}^{\text{pop}}$ satisfies $|\beta_{j,C}^{\text{pop}}|/\text{SE}_T(\beta_{j,C}^{\text{pop}}) \rightarrow \infty$ at rate at least $\sqrt{T}$, with margin $|\beta_{j,C}^{\text{pop}}|/\text{SE}_T \geq \tau + c^*$ for all sufficiently large T .

(A5) Penalty rate. The regularization strength satisfies $\lambda_T \rightarrow 0$ and $\sqrt{T} \lambda_T \rightarrow \infty$ as $T \rightarrow \infty$. The adaptive exponent is $\gamma > 0$.

(A6) Marginal sign agreement on the support. For every discovered cluster $\hat{C}$ and every predictor j with $S_{\text{true},kj} \neq 0$ for some $k \in \hat{C}$, the population cluster-pooled slope satisfies $\text{sign}(\beta_{j,\hat{C}}^{\text{pop}}) = S_{\text{true},kj}$ for every such k . The condition is substantive because the pooled slope is a *marginal* quantity. With correlated predictors, $\beta_{j,C}^{\text{pop}} \propto \sum_l (\sum_{k \in C} \beta_{\text{true},kl}) \Sigma_{F,jl}$, so cross-predictor leakage can flip the marginal sign away from the *partial*

sign that $\mathcal{S}_{\text{true}}$ encodes. (A6) holds when the predictors are uncorrelated, and more generally when the cross-predictor leakage term does not dominate the own-predictor term. The mixed-sign regimes of Section 4 probe the violations of (A3) and (A6).

Statement. Define the support-sign event

$$\mathcal{H}_T = \left\{ \hat{\mathcal{S}}_{T,kj} = \mathcal{S}_{\text{true},kj} \text{ for all } (k,j) \text{ with } \mathcal{S}_{\text{true},kj} \neq 0 \right\}, \quad (11)$$

the event that the gate neither pins a true-support cell to zero nor assigns it the wrong half-space.

Proposition 1 (support-sign and support consistency). Under conditions (A1)–(A6), $\Pr(\mathcal{H}_T) \rightarrow 1$ as $T \rightarrow \infty$, and, conditional on this event, the constrained adaptive sparse group LASSO estimator $\hat{\beta}_T$ recovers the true support, $\Pr(\text{supp}(\hat{\beta}_T) = \text{supp}(\beta_{\text{true}})) \rightarrow 1$.

Remark (truly-zero cells and the fixed threshold). Proposition 1 does not claim $\Pr(\hat{\mathcal{S}}_T = \mathcal{S}_{\text{true}}) \rightarrow 1$. At a fixed threshold τ , the stronger claim is false. On a cell where the population pooled slope is exactly zero, the per-response gate t statistic is asymptotically standard normal, so the per-response gate retains the cell with limiting probability $2\Phi(-\tau) \approx 0.45$ at the default $\tau = 0.75$. The cluster-pooled gate retains such a cell with higher probability when the cluster's responses are correlated, because the pooled variance estimate of Equation (10) sums the $|C|$ per-response residuals and does not adjust for the positive within-cluster correlation that pooling exploits, which over-disperses the pooled null t statistic. The higher pooled false-sign rate in Section 4 is this effect. On a cell where the partial coefficient is zero but the marginal pooled slope is not, through cross-predictor leakage, the gate retains the cell with probability tending to one. Both outcomes leave support recovery to the penalty. The retained constraint is a one-sided half-space whose boundary contains zero, so it never excludes the value $\hat{\beta}_{kj} = 0$ that the adaptive penalty drives the cell toward. A binding half-space on a truly-zero cell clips the unconstrained estimate's sampling noise on one side, a bounded bias of order the cell's standard error rather than a sign error. Exact recovery of $\mathcal{S}_{\text{true}}$ would additionally require a diverging threshold, $\tau_T \rightarrow \infty$ with $\tau_T = o(\sqrt{T})$, together with zero marginal leakage on the off-support cells. We keep the fixed finite-sample default and treat the gate as a screen rather than a model-selection oracle.

Conjecture 1 (oracle distribution). Under the same conditions, the restriction of $\sqrt{T}(\hat{\beta}_T - \beta_{\text{true}})$ to the true support $\text{supp}(\beta_{\text{true}})$ converges in distribution to a centered Gaussian with the same asymptotic covariance as the OLS estimator restricted to the true support. Establishing this requires verifying that the row-aggregated adaptive

weights of the companion paper (Sepp and Kastenholz, 2026a) inherit the divergence-on-zeros and boundedness-on-support rates of the Wang and Leng (2008) argument in the multi-response, sign-constrained setting. We do not prove it here.

Pooling rate. The separation in (A4) is stated at the minimal $\sqrt{T}$ rate, but pooling sharpens it. The gate statistic uses the closed-form standard error of Equation (10), which divides by $\sqrt{|C| \mathbf{x}_j^\top \mathbf{x}_j}$ and so pools the $|C|$ cluster responses with the T observations. For a cluster of size m with independent within-cluster residuals the effective sample size is mT , so the support statistic diverges at the $\sqrt{mT}$ rate, a factor $\sqrt{m}$ above the per-response $\sqrt{T}$ rate. Within-cluster residual correlation lowers the gain. For a common residual correlation $\bar{\rho}$ the design effect $1 + (m - 1)\bar{\rho}$, the same factor that governs a clustered-sample mean, reduces the effective sample to $mT/(1 + (m - 1)\bar{\rho})$ and the realized rate to $\sqrt{m/(1 + (m - 1)\bar{\rho})}$. The realized rate equals $\sqrt{m}$ as $\bar{\rho} \rightarrow 0$ and tends to one as $\bar{\rho} \rightarrow 1$, so the gain is largest as $\bar{\rho} \rightarrow 0$. This residual correlation is distinct from the total correlation that the clustering reads, so a coherent cluster need not carry a large $\bar{\rho}$. Equation (10) estimates the variance as if $\bar{\rho} = 0$, which both overstates the rate above and over-disperses the null statistic, the source of the elevated false-sign rate of Section 4. The divergence and hence Proposition 1 hold for any $\bar{\rho} < 1$. Figure 3 plots the $\sqrt{mT}$ ideal, which the realized recovery approaches from below.

Proof sketch. The argument proceeds in four steps that mirror the layered mechanism. Steps 1–3 establish $\Pr(\mathcal{H}_T) \rightarrow 1$. Step 4 sketches the conditional support recovery and isolates what remains open for Conjecture 1.

Step 1 (slope consistency). Under conditions (A1)–(A2), the cluster-pooled univariate slope $\hat{\beta}_{j,C}$ of Equation (3) satisfies $\hat{\beta}_{j,C} \rightarrow \beta_{j,C}^{\text{pop}}$ in probability as $T \rightarrow \infty$ for every cluster C and predictor j . The convergence applies the law of large numbers to the no-intercept OLS estimator, using the sub-Gaussian tail bound on ϵ to control the deviation rate.

Step 2 (no gating on the support). The closed-form SSR identity of Equation (4) produces a consistent variance estimator $\hat{\sigma}_j^2 \rightarrow \sigma_{j,\text{pop}}^2$ in probability. For every (j, C) pair on the support, the separation condition (A4) gives $|t_{j,C}| \rightarrow \infty$ in probability at rate $\sqrt{T}$, so $\Pr(|t_{j,C}| \geq \tau) \rightarrow 1$ and the probability that the gate pins any true-support cell to zero vanishes. The step makes no claim about off-support cells. At a fixed τ the gate retains an off-support cell with non-vanishing probability, as the Remark notes, and the argument does not need it to abstain.

Step 3 (correct signs on the support). Steps 1 and 2 imply that for every cluster $\hat{C}$ discovered by HCGL and every support predictor j , the derived sign $\text{sign}(\hat{\beta}_{j,\hat{C}})$ converges in probability to $\text{sign}(\beta_{j,\hat{C}}^{\text{pop}})$. The marginal sign agreement condition (A6) equates this population sign with the common true entry $S_{\text{true},kj}$, which is single-valued within the cluster by the coherence condition (A3). The broadcasted sign therefore matches $S_{\text{true},kj}$ for every $k \in \hat{C}$ on the support, and $\Pr(\mathcal{H}_T) \rightarrow 1$ follows from a union bound over the finitely many (j, C) pairs.

Step 4 (constrained adaptive support recovery). Condition on the event $\mathcal{H}_T$, which has probability tending to one by Step 3. On this event, every true-support cell carries its correct half-space constraint, and the support-restricted

estimator is consistent by standard arguments, so the constraint is inactive in the limit and the estimator coincides with the unconstrained adaptive group LASSO on the support. Off-support cells carry either a zero-pin, which enforces the truth directly, or a one-sided half-space containing zero, which cannot exclude the target value and can bind only by pushing the estimate toward zero. Under condition (A5), the adaptive reweighting of Equation (8) with $\gamma > 0$ produces weights that diverge on truly-zero cells and remain bounded on truly-nonzero cells, which yields support recovery by the standard adaptive-penalty argument. This establishes the support-recovery claim of Proposition 1. The distributional claim of Conjecture 1 would require verifying that the row-aggregated adaptive weights of the companion paper (Sepp and Kastenholz, 2026a), an RMS aggregation across the non-pinned predictors of a response rather than a single per-coefficient weight as in the original Wang and Leng (2008) setup, inherit the rate conditions their central-limit argument needs in the multi-response, sign-constrained problem. We leave this to future work.

Remarks. The cluster sign coherence condition (A3) and the marginal sign agreement condition (A6) are the substantive sufficient conditions for the proof. (A3) is weaker than the perfect-cluster-recovery condition that would require $\hat{C}_g = C_g^*$ for every cluster. It permits HCGL to discover a partition that differs from the true clusters at the identity level, as long as the discovered clusters respect the within-cluster sign structure. (A6) is the price of deriving signs from marginal rather than partial evidence, and it is the condition a practitioner should question first when the predictors carry strong cross-predictor correlation of opposing sign. The application of Section 5 shows that HCGL satisfies the coherence condition reliably even when the adjusted Rand index against a reference grouping is moderate.

C. Computation and reproducibility

The method is implemented in the open-source Python package `factorlasso` (Sepp and Kastenholz, 2026b), which expresses the sign-constrained sparse group LASSO of Equation (7) as a convex cone program. The simulation study of Section 4 is reproduced by the scripts `sign_pooling_simulation.py` and `exhibits.py`, which regenerate Tables 1–4 and Figures 1–5 from a fixed random seed. The eQTL application of Section 5 is reproduced by `eqtl_pipeline.py` and `eqtl_exhibits.py`, which regenerate Table 5 and Figures 6–8. The yeast eQTL data are public (Brem and Kruglyak, 2005) and are distributed with the `trigger` Bioconductor package. The reproduction bundle is openly archived with a permanent identifier at <https://doi.org/10.5281/zenodo.21000294>, and developed at https://github.com/ArturSepp/factorlasso/tree/main/papers/sign_pooling_2026. A pinned `requirements.txt` fixes every dependency, including `factorlasso` version 0.7.2, so the reported tables and figures regenerate from the fixed seed under the recorded environment.

Declaration of competing interest

Both authors are employed by LGT Bank. The bank had no role in the design of the method, the analysis, or the decision to submit the work for publication. The authors declare no other competing interest.

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Data availability

The yeast eQTL data are publicly available (Brem and Kruglyak, 2005). The simulated data and the code that regenerates every table and figure are openly available in the reproduction bundle at <https://doi.org/10.5281/zenodo.21000294>, described in Appendix C.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic's Claude Opus 4.8. The model was used to assist with the design and development of the analysis code (including the design of the implementation, drafting code, and reviewing and debugging it) and, additionally, to draft and edit prose. After using this tool the authors reviewed and edited all content and code as needed and take full responsibility for the content of the publication.

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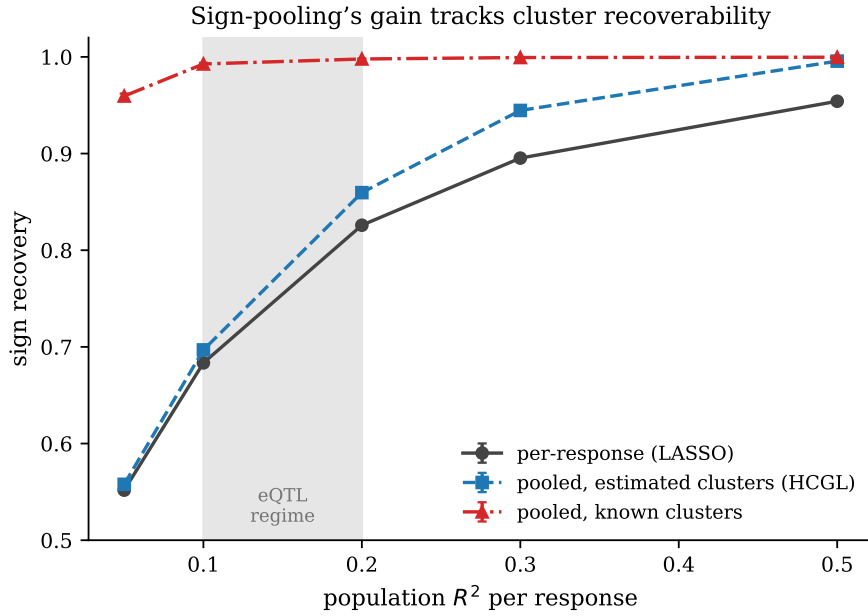


Figure 1: Sign-pooling's gain over per-response derivation tracks cluster recoverability. Sign recovery against population R^2 at fixed $T = 80$, for per-response derivation, estimated-cluster pooling (HCGL), and known-cluster pooling. The shaded band marks the R^2 range of the eQTL application of Section 5. Points are means over 200 replications, and error bars show one standard error.

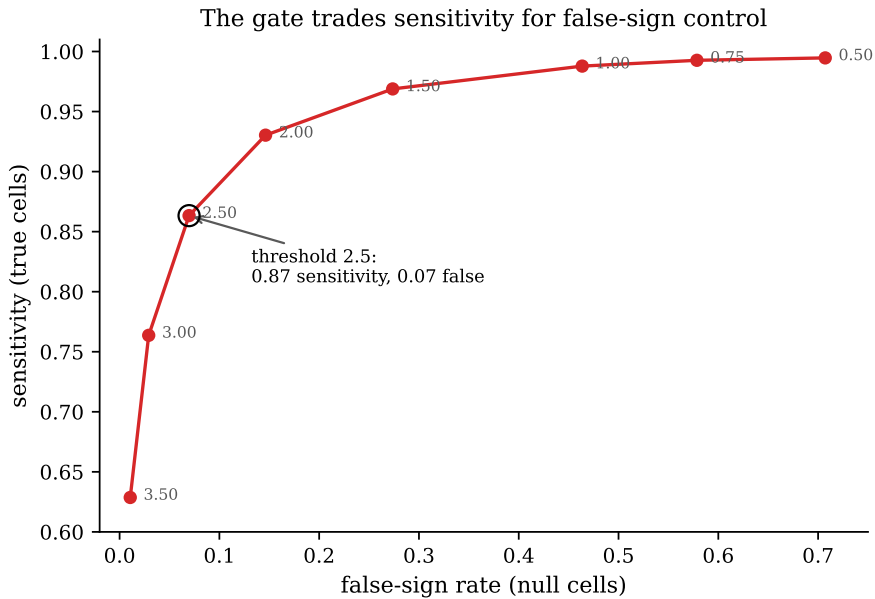


Figure 2: The gate trades sensitivity for false-sign control along a smooth frontier. Sensitivity (recovery on true signals) against the false-sign rate (on null cells) as the threshold τ varies from 0.5 to 3.5, under known-cluster pooling in the base configuration. Labels give τ , and the circled point ($\tau = 2.5$) marks the operating range. Means over 200 replications.

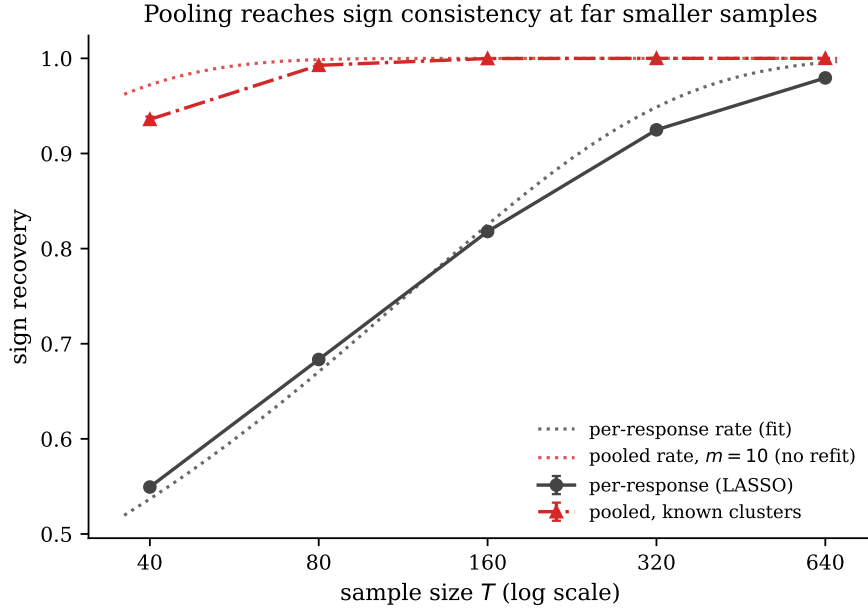


Figure 3: Pooling reaches sign consistency at far smaller samples. Sign recovery against sample size T (log scale) at fixed $R^2 = 0.10$, for per-response derivation and known-cluster pooling. Both rates approach one, and pooling reaches it sooner. Means over 200 replications, and error bars show one standard error. The dotted curves are the Gaussian-gate rate $\Phi(c\sqrt{T} - \tau)$ implied by Appendix B: the constant c is fit to the per-response points, and the pooled curve applies the same c at the cluster-size sample scaling mT ($m = 10$) with no further fit. The ideal leads the data because the mT scaling assumes a homogeneous within-cluster effect, while the simulated loadings are heterogeneous (magnitudes drawn from $U[0.5, 1.5]$), which lowers the pooled effective sample below mT even at the near-zero residual correlation of the base design (Appendix B).

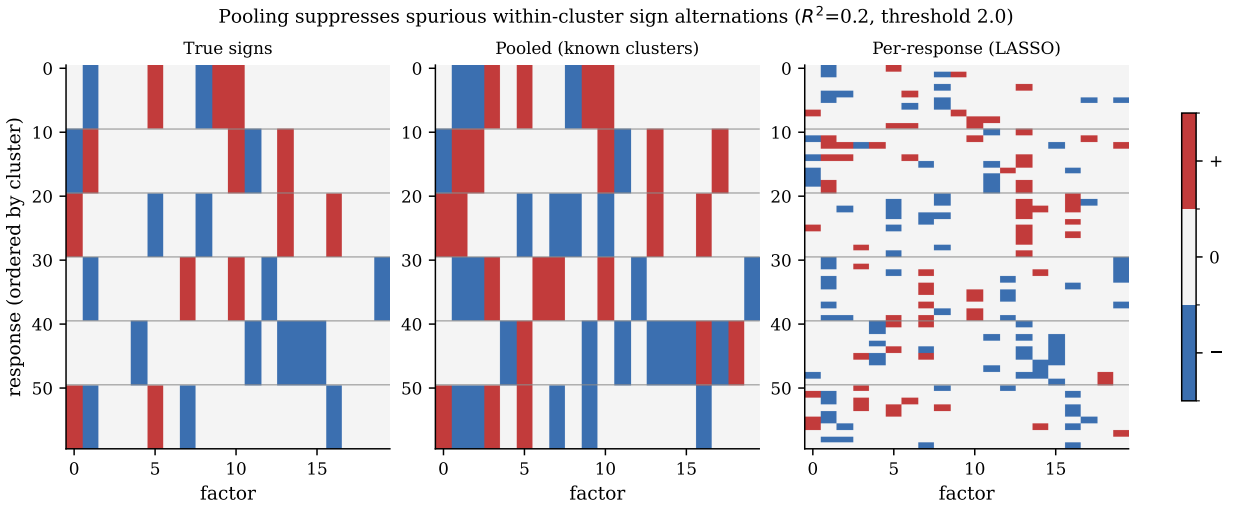


Figure 4: Pooling suppresses spurious within-cluster sign alternations. Population signs (left), known-cluster-pooled signs (center), and per-response signs (right) for one replication at $R^2 = 0.20$ and $\tau = 2.0$. Rows are responses ordered by cluster, and columns are predictors. Blue, white, and red denote -1 , 0 , and $+1$. Horizontal lines separate clusters.

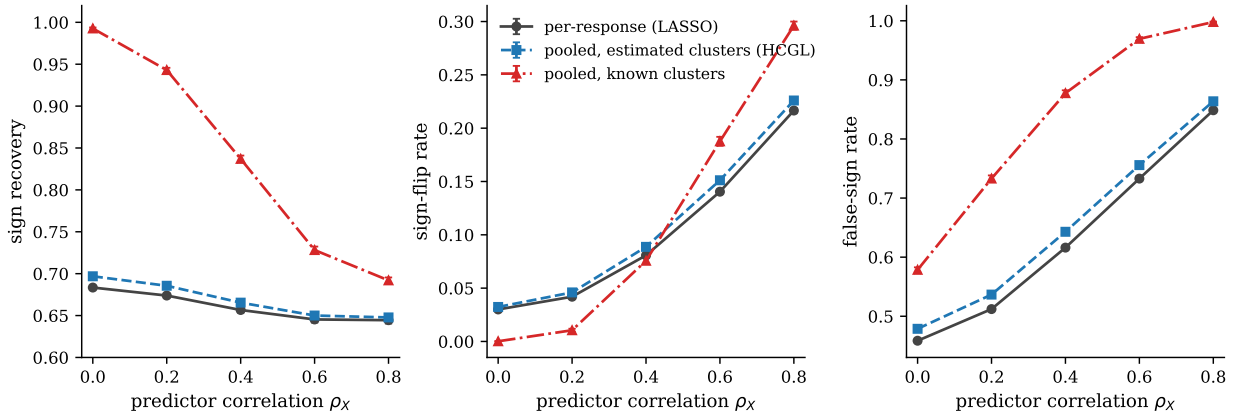


Figure 5: Cluster-pooled sign derivation degrades under correlated predictors. Sign recovery (left), sign-flip rate (center), and false-sign rate (right) against the predictor equicorrelation ρ_X , for per-response derivation, estimated-cluster pooling (HCGL), and known-cluster pooling, in the base configuration ($N = 60$, $M = 20$, $G = 6$, $T = 80$, $R^2 = 0.10$, $\tau = 0.75$). Predictors are drawn with equicorrelation ρ_X , which makes the marginal pooled slope disagree in sign with the partial coefficient. Means over 200 replications, and error bars show one standard error.

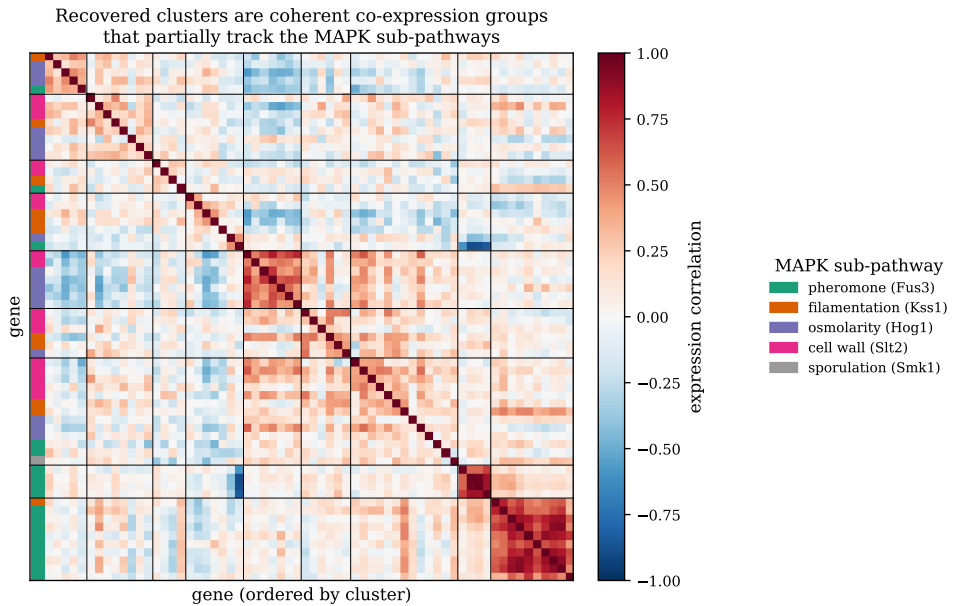


Figure 6: Recovered clusters are coherent co-expression groups that partially track the MAPK sub-pathways. Expression correlation among the 64 MAPK genes, ordered by discovered cluster, with the sub-pathway of each gene shown in the left bar. Black lines separate clusters. Yeast eQTL data of Brem and Kruglyak (2005).

Cluster-pooled signs are coherent within recovered gene clusters (within-cluster agreement 0.93)

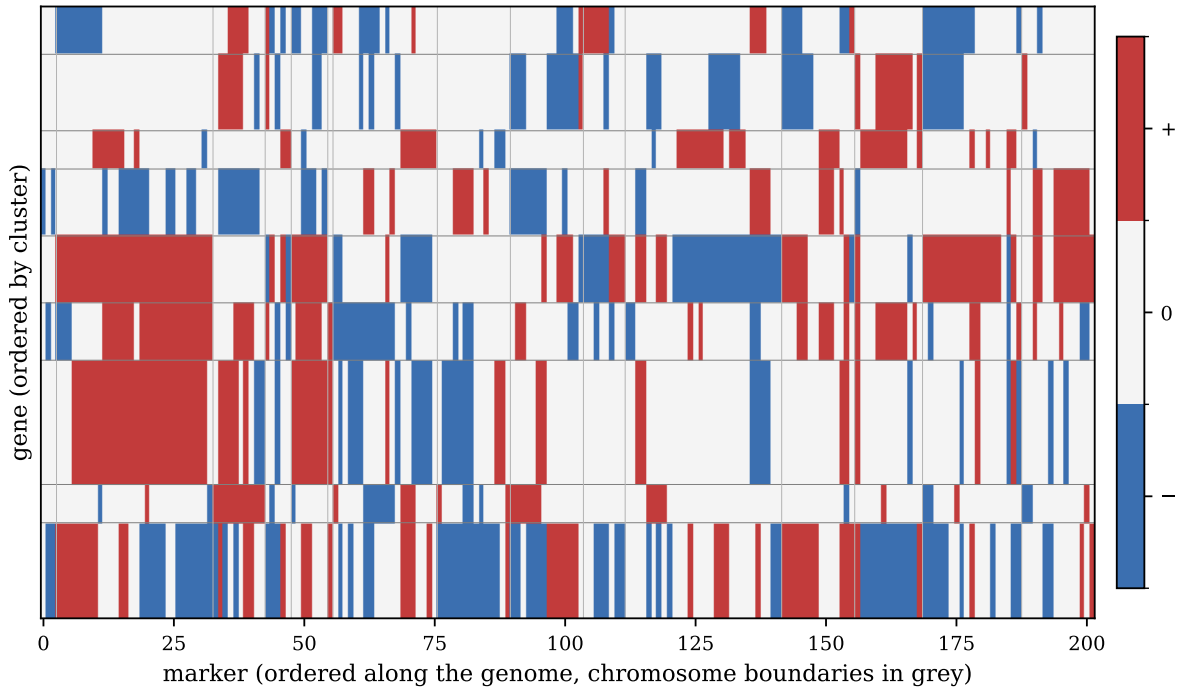


Figure 7: Cluster-pooled signs are coherent within recovered gene clusters. Gated sign matrix for the 64 genes (rows, ordered by cluster) against the 202 markers (columns, ordered along the genome). Blue, white, and red denote -1 , 0 , and $+1$, and grey lines mark chromosome boundaries. The within-cluster sign agreement is 0.93.

A few loci regulate many genes — eQTL hotspots

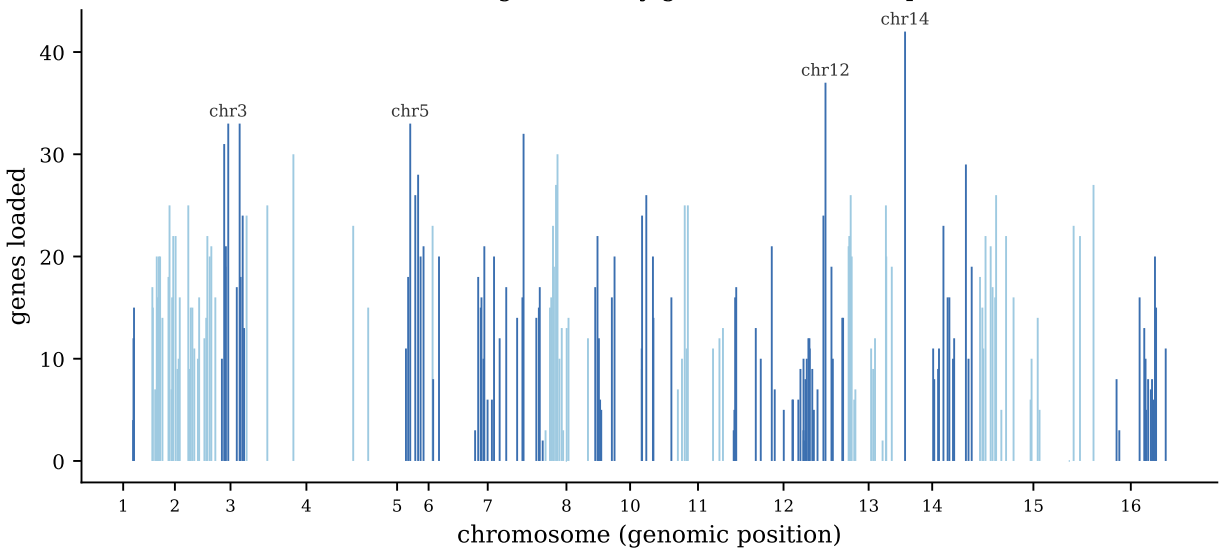


Figure 8: A few loci regulate many genes, the eQTL hotspots. Number of genes loading on each marker against genomic position, for the 202 screened markers. Alternating shades separate chromosomes, and the top four loci are labeled. Yeast eQTL data of Brem and Kruglyak (2005).